

cs5800-hw1_eexercises

Mingxi Zhu

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1 Exercise 0.1

- (a) $f = \Theta(g)$, omit constants, $n^1 = n^1$
- (b) $f = O(g)$, given n^a, n^b , if $a < b$, then $O(n^a) < O(n^b)$
- (c) $f = O(g)$, omit n^1 , $\log n < (\log n)^2$
- (d) $f = \Theta(g)$, omit multiplicative constants, both sides are $O(n \log n)$
- (e) $f = \Theta(g)$, omit multiplicative constants, both sides are $O(\log n)$
- (f) $f = \Theta(g)$, make $n = 10^n$, two sides are $10n$ and $2n$, equals to each other.
- (g) $f = \Omega(g)$, cancel out n , then $n^{0.01} > \log^2 n$, polynomial dominates logarithm
- (h) $f = \Omega(g)$, both sides multiply by $\log n$, cancel out n , then $n > (\log n)^3$, polynomial dominates logarithm
- (i) $f = \Omega(g)$, polynomial dominates logarithm
- (j) $f = O(g)$, both sides multiply by $\log n$, then $(\log n)^{\log n + 1} < n$, polynomial dominates logarithm
- (k) $f = \Omega(g)$, polynomial dominates logarithm
- (l) $f = O(g)$, make $n = 2^n$, then $2^{n/2} < 5^n$
- (m) $f = O(g)$, omit n , base number make huge difference in exponential
- (n) $f = \Theta(g)$, $2^{n+1} = 2^n \cdot 2$, cancel out 2^n
- (o) $f = \Omega(g)$, factorial dominates exponential
- (p) $f = O(g)$, make $n = \log n$, then $n^n / 2^{n^2} = n \log n / n^2$, $n \log n < n^2$
- (q) $f = \Theta(g)$, $\sum_{i=1}^n i^k = (1 + n^k) \cdot n / (k + 1)$, omit lower level polynomial and multiplicative constants, $n^{k+1} / (k + 1) - n / (k + 1) = n^{k+1}$

2 Exercise 0.2

- (1)
if $0 < c < 1$, then $c^{n-1} > c^{n-1} \cdot c$, means $c^{n-1} > c^n$
therefore, $1 > c^1 > c^2 > \dots > c^n$
therefore, 1 is dominant
therefore, (a) is correct.

- (2)
if $c = 1$, then $c^{n-1} = c^{n-1} * c = c^n = 1$

therefore, $1 + c^1 + c^2 + \dots + c^n = \sum_{i=1}^n 1 = n$
therefore, (b) is correct.

(3)
if $c > 1$, then $c^{n-1} < c^{n-1} \cdot c$, means $c^{n-1} < c^n$
therefore, $1 < c^1 < c^2 < \dots < c^n$
therefore, c^n is dominant.
therefore, (c) is correct.

3 Exercise 0.4

(a)
Assume $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $N = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$
 $M \cdot N = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}$
total 4 additions and 8 multiplications.
For X^n , it takes $(4 \cdot 2^n - 4)$ additions and 2^{2n-1} multiplications

(b)
When $n \% 2 = 0$, $X^n = X^{n/2} \cdot X^{n/2}$
When $n \% 2 = 1$, $X^n = X^{n/2} \cdot X^{n/2} \cdot X$
Keep dividing until $n = 1$
total time complexity is $O(\log n)$

4 LLM Disclosure

Use LLM to study the grammar of LaTeX, including how to represent matrix with `\packageamsmath`, etc.